

MUHAMMAD AZIS SAPUTRA

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A fresh graduate in Information Technology from Universitas Sumatera Utara with experience in developing AI-powered solutions, web-based information systems, and data-driven applications. Co-authored a Scopus-indexed publication and gained hands-on experience in computer vision research and machine learning deployment. Interested in leveraging technology, data, and cross-functional collaboration to support impactful product, project, and business initiatives.

Experience & Projects

Plant Biotechnology Laboratory - Medan, Sumatera Utara

May 2025 - Dec 2025

Research Assistant

A research lab at Universitas Sumatera Utara specializing in plant biotechnology and crop improvement.

- Developed a DeepLabV3+ semantic segmentation model with a MobileNetV2 pretrained backbone to support image-based estimation of soybean 100-seed weight as part of a BIMA-funded soybean breeding research project.
- Developed and benchmarked YOLO11s, YOLO12s, and YOLO26s on 1,176 images, selecting YOLO26s for deployment after achieving the highest mAP@0.5:0.95 of 92.01%, with 16.63 ms latency and 60.13 FPS.
- Integrated YOLO26s into a Flask-based web application for batch and real-time soybean seed detection and counting, which was directly validated and approved for research use by the project supervisor
- Co-authored a peer-reviewed research article published in Data in Brief, a Scopus Q3-indexed international journal.

Self-Employed - Medan, Sumatera Utara

Jan 2025 - Apr 2025

Freelance Web Developer

Delivered full-lifecycle custom web development solutions for freelance clients.

- Gathered and analyzed client requirements, translating business needs into functional specifications for two custom web applications.
- Designed, developed, tested, and delivered two web applications, including a project management system and an agricultural equipment e-commerce platform.
- Managed project timelines, client communication, and scope-limited revisions independently, delivering each application according to agreed requirements and schedules.

Bangkit Academy - Indonesia

Sep 2024 - Jan 2025

Machine Learning Cohort

Bangkit Academy 2025 By Google, GoTo, Tokopedia, Traveloka - Machine Learning Cohort

- Selected as one of 4,636 participants from over 45,000 applicants for Bangkit Academy.
- Contributed to the development of Brofin, a machine learning-powered financial planning application featuring 50-30-20 budgeting, house price and inflation prediction, mortgage recommendations, and budget-based product recommendations.
- Collaborated in a seven-member cross-functional team across Machine Learning, Cloud Computing, and Mobile Development to complete and integrate an end-to-end Android application using Agile practices.
- Ranked among the Top 50 teams in the Bangkit Elevator Pitch Challenge.

Indonesian Oil Palm Research Institute (IOPRI) - Medan, Sumatera Utara

Mar 2024 - Jun 2024

Web Developer

Indonesian Oil Palm Research Institute (IOPRI) is a national palm oil center of excellence striving for international research standards.

- Designed and delivered two web-based information systems to digitize internship administration and library operations at IOPRI.
- Developed SIMAGANG, an internship management system supporting participant registration, administrative processing, activity monitoring, and performance assessment.
- Built SIBUKU, a library management system for book inventory, sales recording, and report generation, including database schema design and backend implementation.

Education Level

Universitas Sumatera Utara - Medan, Sumatera Utara

Aug 2021 - Jun 2026

Bachelor of Computer Science, 3.86/4.00

- Dean's List Award Recipient Faculty of Computer Science and Information Technology
- Recipient of Intensive Funding Student Creativity Program in Futuristic Written Ideas (PKM-GFT)
- National Delegate Latihan Mujtahid Dakwah 221: Leadership in Action Summit
- Finalist Outstanding Student Faculty of Computer Science and Information Technology
- Selected for the Final Funding Stage Student Entrepreneurship Program (PMW) USU
- Silver Medalist Invensty National Poster Competition

Organisational Experience

SGC USU - Medan, Sumatera Utara

Jan 2024 - Feb 2025

Head of Innovate Hub Division

Startup Generation Community (SGC) USU is a student organization dedicated to startup development and innovation.

- Led a 10-member Innovate Hub team, planning, coordinating, and overseeing the division's annual work programs.
- Initiated and managed three flagship programs, EduHub, InHub, and ExHub, focused on startup education, idea development, and solution experimentation.
- Coordinated collaboration across divisions, delegated tasks, and monitored progress to ensure programs were delivered according to schedule.

MPMF Fasilkom - TI USU - Medan, Sumatera Utara

Aug 2023 - Dec 2023

Chairperson

Majelis Permusyawaratan Mahasiswa Fakultas (MPMF) is the student legislative body ensuring accountability and policy development at Fasilkom-TI USU.

- Led a 20-member student legislative council, coordinating committee activities, council meetings, and official deliberations.
- Oversaw the implementation and performance of work programs conducted by the Faculty Student Executive Board, supporting accountability across the faculty student government.
- Led the drafting, review, and ratification of the Faculty Student Constitution, covering student rights, organizational governance, and election procedures for future administrations.

Mengajar di Desa by SGC USU - Ndeskati, Sumatera Utara

Aug 2023 - Dec 2023

Secretary of Creative Media Division

Mengajar di Desa is an SGC USU initiative empowering rural communities through educational volunteerism.

- Supported the Creative Media Division in planning communication activities, documenting meetings, and coordinating internal information flow.
- Designed one reusable visual template to introduce the program's organizational structure and members.
- Developed and produced Instagram Story content directly during on-site educational activities to document the program and support public engagement.

Rumah Kepemimpinan Regional 6 Medan - Medan, Sumatera Utara

Aug 2022 - Dec 2022

Head of Communications and Informatics Division

Rumah Kepemimpinan is a national scholarship and mentoring program preparing students for impactful leadership.

- Led a 3-member Communications and Informatics Division, assigning design tasks, reviewing deliverables, and coordinating visual communication initiatives.
- Produced approximately 50 Instagram feed posts at an average frequency of three posts per week, supported by more than 10 reusable templates for recurring organizational programs.
- Published approximately 100 Instagram Stories to provide live coverage of organizational activities and communicate program updates to the audience.
- Developed the official cabinet logo, visual identity guidelines, and promotional materials introducing the cabinet structure and organizational vision.

Skills, Achievements & Other Experience

- **Achievements** 🏆: Dean's List Award Fasilkom-TI Recipient; PKM-GFT Funding Recipient; Outstanding Student Fasilkom-TI Finalist; PMW USU Final Funding Participant; National Delegate LMD 221; Inventsy National Poster Competition Silver Medalist.
- **Hard Skills**: Artificial Intelligence (Machine Learning, TensorFlow, PyTorch, Computer Vision, YOLO), Data Analytics (Goggle Sheet, Excel, Python, SQL, Data Cleaning, Data Preprocessing, Exploratory Data Analysis, Data Visualization), Software Engineering (PHP, JavaScript, Yii, Laravel, React, Next JS, Flask, REST API, Model Integration, Object-Oriented Programming, MVC Architecture, MySQL), Research & Scientific Computing (Experimental Design, Model Evaluation, Scientific Writing).
- **Soft Skills**: Leadership, Analytical Thinking, Problem Solving, Cross-functional Collaboration, Project Management, Communication & Public Speaking.
- **Projects**: Website (supersoybean.cloud), Mobile App (brofin)
- **Language**: English (Intermediate), Indonesia (Native)